Preliminary LOX/CH4 Rocket Engine Performance and Thermal Analysis

Preliminary propulsion engineering study

LOX / CH4 | 30 bar chamber pressure | 20 kN conceptual thrust

Executive Summary

This report presents a preliminary 20 kN liquid rocket engine concept using LOX and methane. The purpose was to connect the main propulsion calculations into one engineering workflow. NASA CEA was used for the thermochemical study, followed by preliminary engine sizing, injector flow-area sizing, a first-order regenerative-cooling check and a new 2-D axisymmetric nozzle CFD case.

The O/F sweep selected O/F = 3.0 for the rest of the study. At 30 bar and an area ratio of 10, CEA gave c* = 1851.6 m/s, Cf = 1.6357 and about 308.8 s specific impulse. The preliminary geometry has a 72.0 mm throat and a 227.8 mm exit.

The Fluent case converged at iteration 1038. It predicted 19.63 kN thrust, 3.0952 exit Mach and 6.5103 kg/s mass flow.

1. Design Basis

The design point was kept simple because the purpose of the project is preliminary propulsion analysis rather than final engine design.

Parameter	Value
Propellants	LOX / CH4
Chamber pressure	30 bar
O/F ratio	3.0
Nozzle expansion ratio	10
Target thrust	20 kN
CH4 reference inlet temperature	111.7 K
LOX reference inlet temperature	90.2 K

2. Thermochemical Performance

NASA CEA was used to study O/F ratios from 2.50 to 3.50. Chamber pressure was fixed at 30 bar and the nozzle area ratio was fixed at 10.

O/F	Tc (K)	c* (m/s)	Cf	Exit Mach	Exit P (kPa)	Isp (s)
2.50	3115.2	1847.4	1.6174	3.300	36.191	304.7
2.75	3277.7	1857.8	1.6271	3.219	38.703	308.3
3.00	3373.8	1851.6	1.6357	3.133	41.726	308.8
3.25	3423.8	1835.4	1.6399	3.064	44.928	306.9
3.50	3445.4	1814.5	1.6401	3.042	47.217	303.4

O/F = 3.0 was selected because it gave the highest reported specific impulse in the tested range. This selection is specific to the conditions studied.

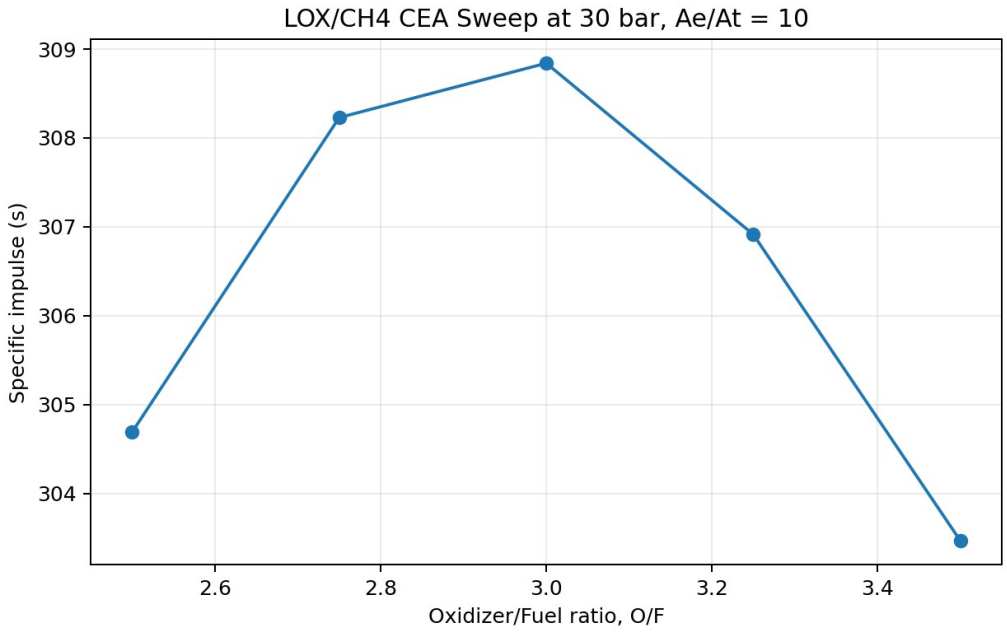


Figure 1. Specific impulse from the CEA O/F sweep.

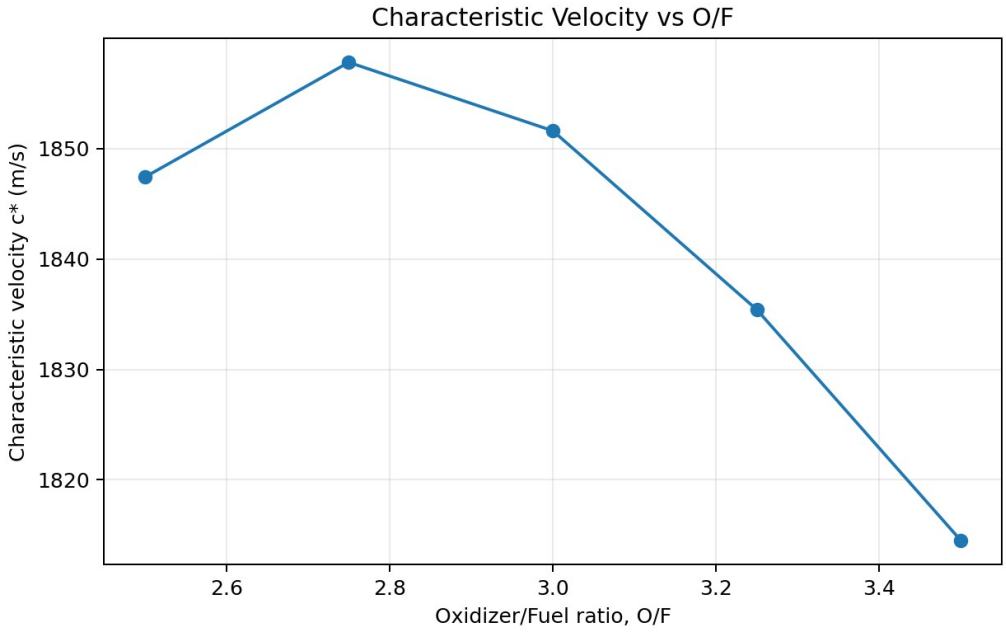


Figure 2. Characteristic velocity from the CEA O/F sweep.

3. Preliminary Engine Sizing

A 20 kN target thrust was used to size the throat. The throat area was obtained from F = Cf Pc At. Total mass flow was then obtained from mdot = Pc At / c*.

Quantity	**Result**
Throat area	40.76 cm²
Throat diameter	72.04 mm
Exit area	407.57 cm²
Exit diameter	227.80 mm
Total mass flow	6.604 kg/s
LOX mass flow	4.953 kg/s
CH4 mass flow	1.651 kg/s

These dimensions form the basis of the new CAD and CFD model. They are not final manufacturing dimensions.

4. Preliminary Injector Sizing

The injector was treated as a simple liquid-orifice flow problem. An injector pressure drop equal to 20% of chamber pressure was assumed.

Quantity	**LOX**	**CH4**
Mass flow	4.953 kg/s	1.651 kg/s
Assumed density	1141 kg/m³	422 kg/m³
Effective flow area	167.3 mm²	91.7 mm²

The assumed pressure drop is 6 bar, giving a nominal injector-side pressure of about 36 bar before other feed-system losses. This is preliminary flow-area sizing only. Atomization, mixing, combustion stability and experimental validation were not studied.

5. Regenerative-Cooling Screening

A simple throat thermal model was used to estimate the scale of the heat load. The model combines CEA gas properties with a screening-level hot-side heat-transfer estimate and a simplified methane coolant-channel model.

Parameter	Result
Throat temperature	3200.3 K
Estimated recovery temperature	about 3371 K
Hot-side coefficient	about 26.3 kW/(m² K)
Coolant-side coefficient	about 60 kW/(m² K)
Estimated heat flux	49.8 MW/m²
Hot-side wall temperature	1478 K
Coolant-side wall temperature	1230 K

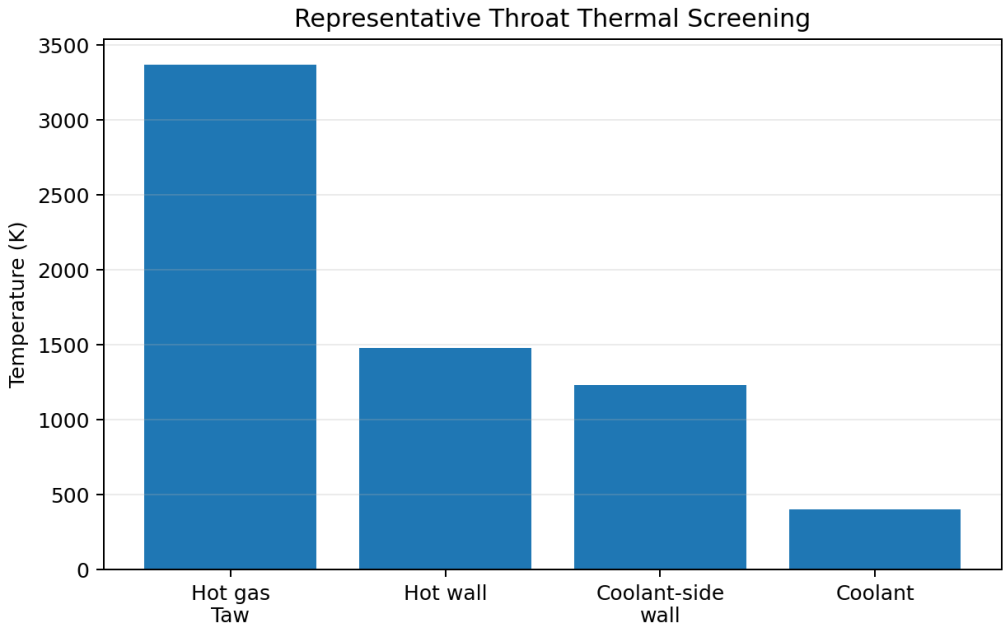


Figure 3. Representative throat thermal screening.

These values are screening results, not a final cooling design. A detailed analysis would require variable-property methane treatment and a more detailed cooling model.

6. Nozzle Geometry and Mesh

A new 2-D axisymmetric nozzle was created from the calculated engine dimensions. The previous arbitrary nozzle geometry was not reused.

Geometry item	Value
Chamber radius	50.0 mm
Chamber length	100.0 mm
Throat diameter	72.04 mm
Exit diameter	227.80 mm
Converging half-angle	10°
Diverging half-angle	12°

The mesh contains 13,193 elements and 13,510 nodes. Maximum skewness is 0.4387 and minimum orthogonal quality is 0.7643. The throat was locally refined.

7. CFD Setup and Results

The Fluent model uses a steady density-based solver, the energy equation and an inviscid flow model. The gas is represented with a constant-property ideal-gas surrogate based on the selected CEA chamber state. The chamber total pressure is 30 bar and the chamber total temperature is 3373.82 K. The exit pressure is 0.41726 bar.

Quantity	Reference	Fluent CFD	Difference
Thrust	20.00 kN	19.63 kN	-1.86%
Mass flow	6.604 kg/s	6.5103 kg/s	-1.41%
Exit Mach	3.133	3.0952	-1.21%
Exit pressure	41.726 kPa	44.915 kPa	7.64%
Exit temperature	2005.2 K	2051.3 K	2.30%

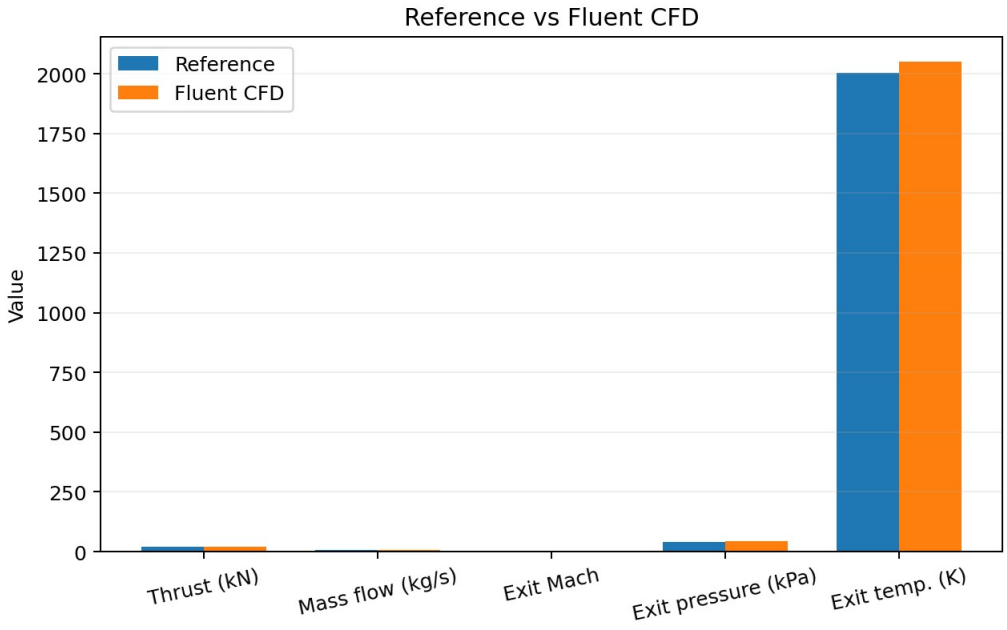


Figure 4. Reference values compared with the converged Fluent case.

The solution converged at iteration 1038. The final inlet and outlet mass-flow magnitudes were 6.5096 and 6.5103 kg/s, giving a mass-flow imbalance of about 0.011%. The CFD exit Mach is within 1.2% of the CEA value, and the CFD thrust is within 1.9% of the 20 kN design target.

The differences are expected. CEA uses equilibrium thermochemistry and variable gas properties, while the Fluent case uses a constant-property ideal-gas approximation. The CFD case is therefore a check on the preliminary nozzle sizing and compressible-flow behavior, not a full chemical-flow reproduction.

8. Limitations

- Detailed injector geometry, atomization and mixing were not modeled.
- Combustion stability was not assessed.
- Turbopump design and rotor dynamics were not included.
- The methane cooling model uses screening-level constant-property assumptions.
- The thermal model is not a full 3-D conjugate heat-transfer or thermal-structural analysis.
- Film cooling and manufacturing constraints were not evaluated.
- No experimental or hot-fire validation was performed.

9. Conclusion

This project connects the main steps of a preliminary liquid rocket propulsion analysis. The CEA sweep provided a practical design point, which was then converted into a 20 kN preliminary engine size. That size was used for injector flow-area sizing, a first thermal check and a new nozzle geometry.

The final Fluent case converged and predicted 19.63 kN thrust with an exit Mach number of 3.095. The result is close to the preliminary target and the CEA reference, while the limitations clearly define what would need higher-fidelity work in a real engine program.

References

- NASA - Chemical Equilibrium with Applications (CEA) and CEARUN documentation.
- NASA Glenn Research Center - Rocket thrust and liquid rocket engine guidance.
- Bartz-type hot-gas heat-transfer correlation for rocket nozzles.